Day: 04

* What is TCP?

- TCP (Transmission Control Protocol) is a connection-oriented, reliable transport layer protocol (Layer 4 of OSI). Before any data exchanged, two hosts must establish a connection. This is done through the 3-Way Handshake.

* TCP Segment Structure

Field	Size	Purpose
Source Port	16 bits	Sender's port number
Destination Port	16 bits	Receiver's port number
Sequence Number	32 bits	Tracks byte order of data
Acknowledgement Number	32 bits	Next expected byte
Flags (Control Bits)	6 bits	SYN, ACK, FIN, RST, PSH, URG
Window Size	16 bits	Flow control - how much data to accept

* The 6 TCP Flags

- SYN
  Synchronise — initiate connection

- ACK
  Acknowledge — confirms receipt

- FIN
  Finish — close connection

- RST
  Reset — abruptly terminate

- PSH
  Push — send data immediately

- URG
  Urgent — priority data

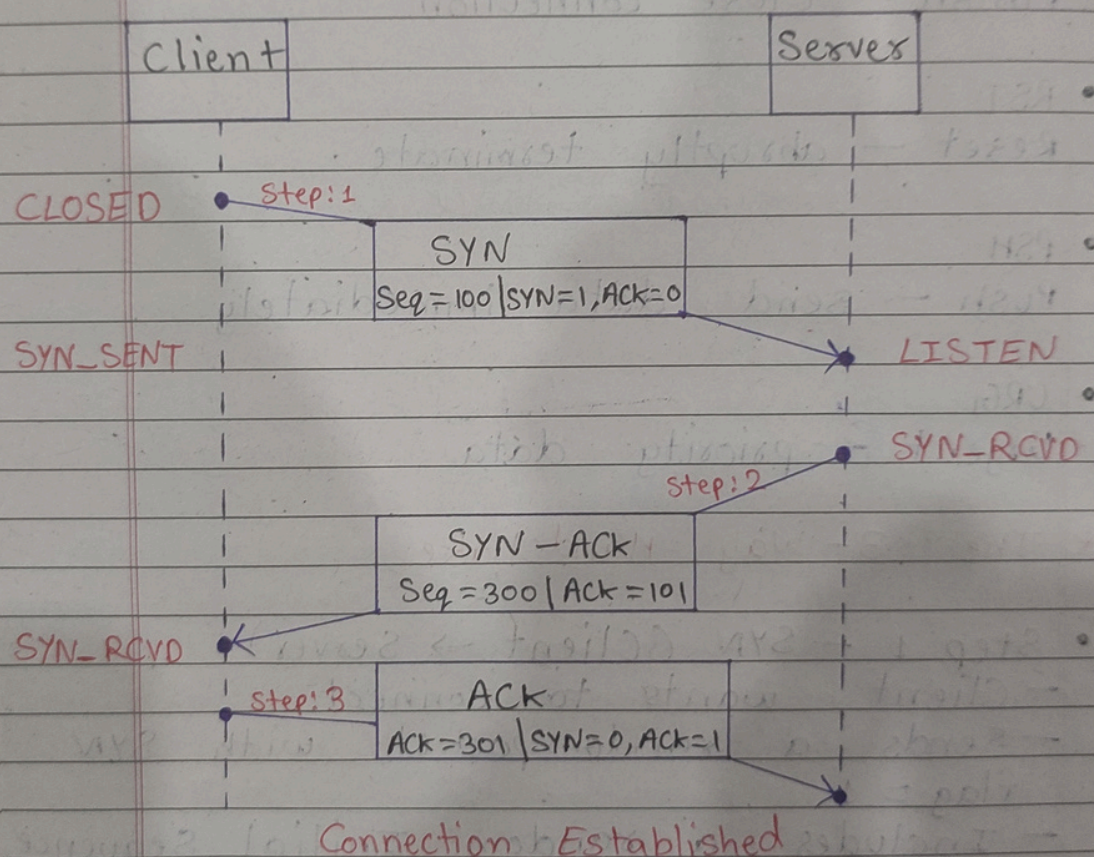
* The 3-Way Handshake

- Step 1 — SYN (Client → Server)
  - Client wants to connect
  - Sends a ~~random~~ packet with SYN flag = 1
  - Includes a random Initial Sequence Number (ISN), e.g. Seq = 100
  - ACK flag = 0 (no acknowledgement yet)

- Step 2 — SYN - ACK (Server → Client)
  - Server accepts the request
  - Sends back SYN = 1, ACK = 1
  - Server's own ISN, eg. Seq = 300
  - Acknowledges client's ISN : ACK number = 101 (Client's ISN + 1)

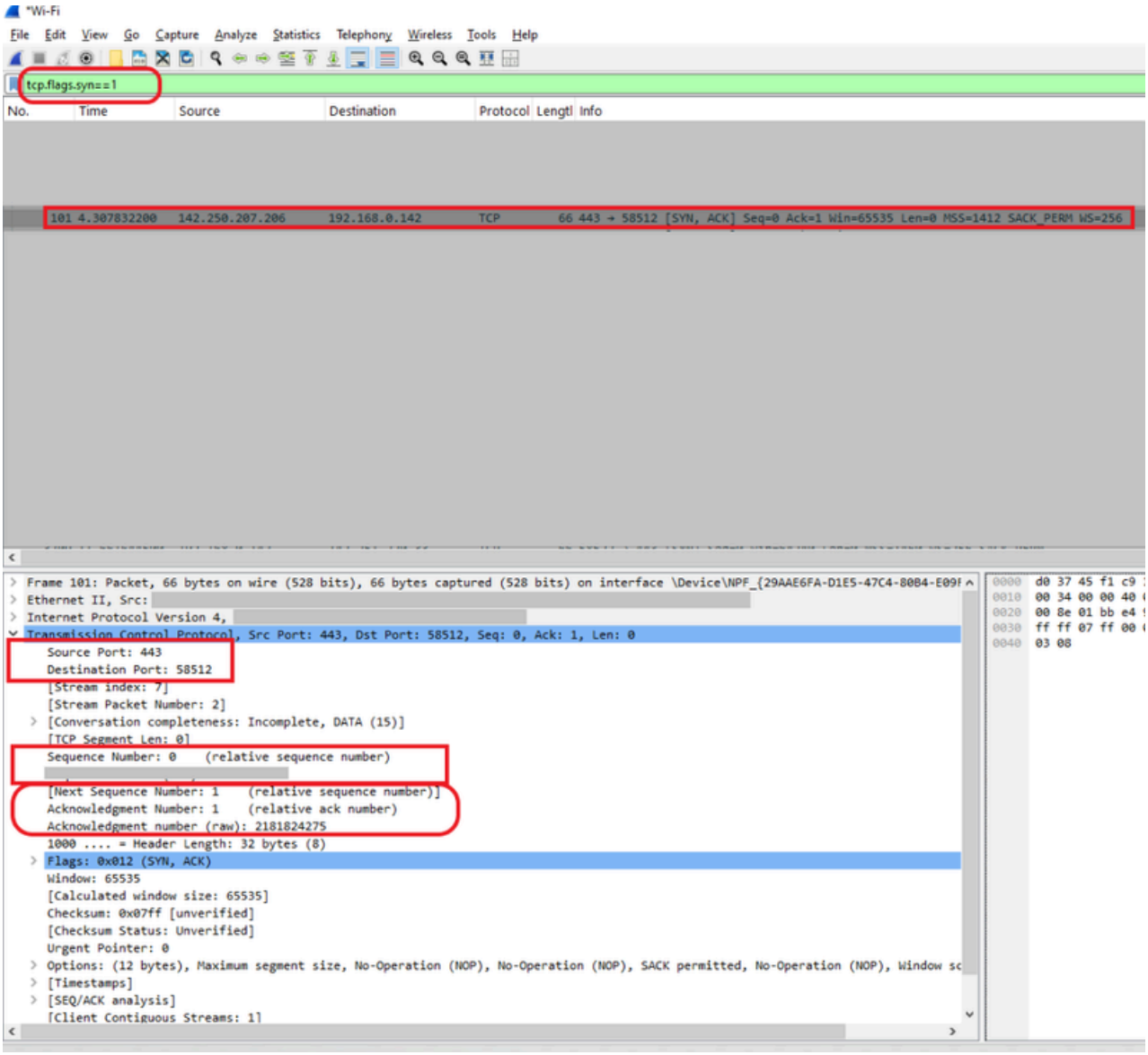
- Step 3 — ACK (Client → Server)
  - Client acknowledges the server
  - Send ACK = 1, SYN = 0
  - ACK number = 301 (Server's ISN + 1)
  - Connection is now <u>Established</u>

* TCP 3-Way Handshake



* What does this matter for DFIR?

- Every malware C2 connection starts with a SYN packet.
- The filter tcp.flags.syn == 1 is one of the most used filters in forensics.


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.flags.syn==1
No. Time Source Destination Protocol Lengtl Info
101 4.307832200 142.250.207.206 192.168.0.142 TCP 66 443 → 58512 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
Frame 101: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{29AAE6FA-D1E5-47C4-80B4-E09F
Ethernet II, Src:
Internet Protocol Version 4,
Transmission Control Protocol, Src Port: 443, Dst Port: 58512, Seq: 0, Ack: 1, Len: 0
Source Port: 443
Destination Port: 58512
[Stream index: 7]
[Stream Packet Number: 2]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 2181824275
1000 .... = Header Length: 32 bytes (8)
Flags: 0x012 (SYN, ACK)
Window: 65535
[Calculated window size: 65535]
Checksum: 0x07ff [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), SACK permitted, No-Operation (NOP), Window sc
[Timestamps]
[SEQ/ACK analysis]
[Client Contiguous Streams: 1]
0000 d0 37 45 f1 c9
0010 00 34 00 00 40
0020 00 8e 01 bb e4
0030 ff ff 07 ff 00
0040 03 08